A11y in Angular Router

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A11y in Angular Router

- Unique page titles
- Active link identification
- Focus management
  - (is it a good idea?)

Unique page titles

- Add unique page titles to your routes
  - dynamic titles can be set via inject(Title)

```
export const appRoutes: Route[] = [
  {
    path: 'home',
    component: HomeComponent,
    title: 'Home - NG A11y'
  },
]
```

Active link identification

– The *RouterLinkActive* directive provides the *ariaCurrentWhenActive* input for aria-current

```
<nav>
  <a routerLink="home"
       routerLinkActive="active-page"
       ariaCurrentWhenActive="page">
       Home
  </a>
  <a routerLink="about"
       routerLinkActive="active-page"
       ariaCurrentWhenActive="page">
       About
  </a>
</nav>
```

Focus management

- Focus management after navigation
  - Avoid relying solely on visual cues by making sure routing updates focus after page navigation
  - Use the *NavigationEnd* event from the NG Router service to know when to update focus

```
inject(Router).events.pipe(
  filter((e) => e instanceof NavigationEnd)
).subscribe(() => {
  this.document.querySelector('main')?.focus();
});
```

Demo

Angular Router

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Angular CDK

- Angular CDK includes the a11y package that provides tools to support various areas of accessibility
  - *LiveAnnouncer* is used to announce messages for screen-reader users using an aria-live region

```
this.liveAnnouncer.announce('Flights loading error');
```

  - *cdkTrapFocus* directive traps Tab focus in an element

```
<div class="gdpr-dialog" cdkTrapFocus>
  <!-- Yay, focus won't leave this element! -->
</div>
```

Demo

LiveAnnouncer Demo

Angular Material

– Angular Material library is a suite of reusable UI components that aims to be fully accessible

– If you're building your own Design System, you can copy what Angular Material is doing ☺

UI Components in NG

- Augmenting native elements
  - Native HTML elements capture several standard interaction patterns that are important to A11y
  - When authoring NG components, you should re-use native elements
  - Instead of creating a custom element for a new variety of button, create a component that uses an attribute selector with <button>
  - This most commonly applies to <button> and <a>, but can be used with many other elements
  - Example: MatButton
- Using containers for native elements
  - Sometimes using the appropriate native element requires a container element
  - E.g., the native <input> element cannot have children, so any custom text entry components need to wrap an <input> with extra elements
  - Example: MatFormField

Demo

Showcase NG Material

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Design Systems

- Canopy
  - https://github.com/Legal-and-General/canopy
- Clarity
  - https://clarity.design/
- Prime NG
  - https://primeng.org/guides/accessibility
- **KoliBri - Public UI**
  - https://public-ui.github.io/docs/components/input-text
- You can still use Angular CDK with all of them ☺

Questions?

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